

	TITLE:			DOCUMENT NUMBER:								
	TASK RISK ASSESSMENT			FC-GL-EHS-MAN-0501-F3 (Tab Q.1)								
WORK ACTIVITY:		PROCEDURE REFERENCE NUMBER		Iss / Rev	TRA No.		Iss / Rev:	Date:				
Using High Powered Hydraulic Bolting/Torquing Tooling		Standard Hazard Specific TRA		N/A	TRA_STD_Hydraulic Bolting/Torquing		1.0	26th Oct 2023				
PERSONNEL IMPACTED (Tick all applicable)												
☒ 1FS Employees ☒ GE Employees ☒ Customer ☒ GE Contractors ☒ 3rd Party Contractor ☐ Public ☐ Others (Specify)												
Special Notes: The TRA covers standard used of high powered torquing equipment (e.g. ITH, Riverhawk, Hytorc) The safe method of work (method statement) should be discussed as part of the Pre Task Brief, any defences and controls that cannot be fully and effectively implemented should initiate a STOP WORK. Dynamic work place hazards should be addressed using the POWRA through the principles of "What, How, What, How" before the work activity commences and when any new hazards are introduced to the work place. Hydraulic bolting/torquing tooling shall only be used by trained and qualified operators. For contractors, check individual training certification records. Any identified defective equipment shall be red tagged and securely quarantined.												
Severity		Frequency		SEVERITY					Risk Rating	Strategy		
S5	Major	F5	Monthly	FREQUENCY	S1	S2	S3	S4	S5	High	Intervention - (Critical Step)	
S4	Severe	F4	Quarterly		F5	5	6	7	8	9	PPE cannot be assigned as a Primary Defense	
S3	Significant	F3	Annually		F4	4	5	6	7	8	Control	
S2	Moderate	F2	Bi-Annually		F3	3	4	5	6	7	Do not proceed - additional defenses required	
S1	Minor	F1	> two years		F2	2	3	4	5	6	Monitor	
					F1	1	2	3	4	5	Low	Maintain Defenses
*Risk = (Severity + Frequency) -1												
TASK STEPS		POTENTIAL RISK (PR)				DEFENCES:		CONTROLLED RISK (CR)				
		HAZARDS:		S	F	PR*			S	F	CR*	Safe Method of Work
1	Pre-Use Inspection of Hydraulic Bolting/Torquing Tool and components	Manual Handling (Ergonomics), Dropped Objects		3	2	4	Laydown Planning, Manual Handling Technique, Training/Competency, PPE		1	1	1	Both Pump and Tool Operators of hydraulic bolting/torquing tools must be fully trained and qualified in the operation of the specific type of tool. Documented equipment-specific Inspection Checklist must be conducted using the following forms: - FC-GL-EHS- SAF-0709-F1 (TabX1): ITH Inspection Checklist. - FC-GL-EHS- SAF-0709-F2 (TabX2): HYTORC/Hi-Force Inspection Checklist. - FC-GL-EHS- SAF-0709-F3 (TabX3): Riverhawk Inspection Checklist. Confirm approved back-up wrenches are available for all through-bolt applications. Ensure the pre-use inspection is conducted in controlled environment away from the point of work. Impact and cut resistant gloves and toe protection footwear shall be worn Use correct body posture when lifting equipment - use lift assistance if needed. Do not use hydraulic hoses, uniswivels, pump power or remote cords as means of moving the equipment. Ensure walkways and path to task location are clear and free of obstructions. Utililse dropped object prevention equipment (e.g. tethers, toolbags etc..) as need identified. If using Forklift to take palletised tooling to work location it must be pre-use inspected and operated by a trained operator Complete POWRA (FC) or STA (APM) per policy identifying all potential
2	Take Hydraulic Bolting/Torquing tool and components to the point of work	Walking/Working Surface, Manual Handling (Ergonomics), Dropped Objects, Power Industrial Trucks		3	3	5	Plan the route, POWRA/POWLA, Dropped Object Prevention Tool Kit, Manual Handling Technique, Training/Competency, Housekeeping Program		1	1	1	

3	Set up Hydraulic Hoses	Pressurised Energy, Walking/Working Surface	2	3	4	Certified Equipment, Pre-use inspection, POWRA/POWLA, PPE, Training/Competency, Whip Arrestors	1	2	2	Hoses and connections should be checked before use and confirm whip arrestors are attached at all the connecting points. Hoses should be run so they are trip hazard free and free from mechanical damage risk. Apply pressure slowly to ensure no leaks or sudden uncoupling. The working pressure must be within the limits for hoses and pump (i.e. the maximum working pressure shall not be exceeded for hoses or pump)
4	Locate Torquing Tool onto bolt head.	Manual Handling (Ergonomics), Dropped Objects, Physical Exertion	3	3	5	Dropped Object Prevention Tool Kit, Manual Handling Technique, Training/Competency, Peer Checking	1	1	1	Use correct body posture when manually handling equipment - get assistance if needed from peer. Ensure safety handle is secured to the tool to avoid pinch points. Ensure the appropriate sized socket is applied to the tool. Place the tool on the nut or bolt assuring that it is fully engaged. Check that the hoses are not in danger of being kinked or damaged.
5	Set-up reaction arm and hands-free backup wrenches (where applicable)	Pinch Points/Line of Fire, High Powered Tooling	3	4	6	Approach/Restriction Boundaries, Communication, PPE, Training/Competency, Supervisory Oversight, Certified Equipment	1	2	2	Ensure the reaction point is secure, the reaction arm must be rotated so that it rests against the adjacent nut in the same plane as the nut to be tightened. Do not use shims between the reaction arm and the equipment. Ensure proper back up wrenches are used on all through bolts.
6	Operate the tool to loosen or tighten bolt.	High Powered Tooling, Dropped Objects, Pressurised Energy, Pinch Points/Line of Fire, Projectiles	3	4	6	Approach/Restriction Boundaries, Communication, PPE, Supervisory Oversight, Training/Competency	2	1	2	Clear controlled and maintained approach boundaries shall be set-up to prevent access to those not involved in the tool operation. Ensure proper body positioning to keep body and face out of Line of Fire. Ensure face shield and hearing protection is worn as well as impact resistant gloves. Full coveralls must be worn to mitigate risk from high pressure hydraulic oil leak. Keep hands and fingers out of pinch points. Be aware of adjacent activities and associated hazards. Pump operator is responsible for advancing the tool and ensuring the Tool Operator is ready to pressurize. Clear communication must be maintained before beginning any torquing operation (for example use, ALL CLEAR & HOLD rather than NO or GO or
7	Remove or Replace Bolt from Bolt hole	Dropped Objects, Manual Handling (Ergonomics), Physical Exertion	3	3	5	Dropped Object Prevention Tool Kit, Manual Handling Technique, Training/Competency, Peer Checking, Designated Storage/Laydown	1	1	1	Use correct body posture when lifting equipment - get assistance if needed from peer. Ensure walkways to/from designated laydown area are clear and free of obstructions. Utilise dropped object prevention equipment (e.g. tethers, toolbags etc..) as need identified. Confirm back-up wrenches have been removed and stored correctly.

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